



Book I. Prop. XV. Theor. *If two right lines (— and —) intersect one another, the vertical angles  and ,  and  are equal.*

$$\text{yellow sector} + \text{red sector} = \text{semicircle}$$

$$\text{black sector} + \text{red sector} = \text{semicircle}$$

$$\therefore \text{yellow sector} = \text{black sector}.$$

In the same manner it may be shown
that

$$\text{red sector} = \text{blue sector}.$$

Q. E. D.